LESSON GUIDE

Technology Plus

Unit 5: Software

3.2.1 – Purposes of Operating Systems

Lesson Overview:

In this lesson, students will identify key functions of operating systems. Then, they will classify different user tasks by the OS function that handles it.

Students will:

- Identify key functions of operating systems.
- Classify user tasks by the OS function that handles it.

Guiding Question: In what ways do operating systems interface between software and hardware?

Suggested Grade Levels: 8-12

Technology Needed: No

Approximate Time Required: 60-80 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 3.2 – Explain the purpose of operating systems.
  - Interface between applications and hardware
  - Disk management
  - Task and process management
  - Application management
  - Device management
  - Access control

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Lesson Background

Background Information:

Most students should be familiar with operating systems, which are the base software installed on their devices (phones, computers, gaming consoles, etc.), but may not know anything about how they work and what they control. This lesson will introduce the primary functions of most operating systems. Subsequent lessons will further explore specific types of operating systems, OS files systems and their management, and OS components.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 3.2.1 – Purposes of OS
Materials per Group	• Purposes of Operating Systems Scenarios
Materials per Student	• Student Notes 3.2.1 – Purposes of Operating Systems • Student Handout 3.2.1 – Purposes of Operating Systems • Activity Slides 3.2.1 – Can You Handle This • Assessment 3.2.1 – Purposes of Operating Systems (optional)

Lesson Background

Cyber Terms:

Term	Definition
operating system (OS)	a set of programs that manage how hardware works, and how that hardware communicates with other programs
hardware abstraction	a layer of an operating system that translates high-level commands and APIs to hardware-specific instructions
system calls	a controlled interface where applications can perform low-level operations by requesting them through the OS and having the OS translate them into hardware actions
graphical user interface (GUI)	a visual system that allows a user to interact with an operating system using icons, buttons, and menus instead of text-based commands
command line interface (CLI)	a text-based system that allows a user to interact with an operating system using text-based commands instead of icons, buttons, and menus
disk management	organizing, managing, and maintaining storage devices
task and process management	overseeing and controlling the execution of tasks, or processes, within a computing environment
application management	overseeing application execution, performance, and interaction within a computing environment
device management	controlling and coordinating hardware devices
access control	regulating who can access what resources and what actions they can perform

3.2.1 – Purposes of Operating Systems

Guiding Question: In what ways do operating systems interface between software and hardware?

Lesson Launch (3-5 minutes)

How Do I Use This Thing?

- Define ***operating system*** (OS) as a set of programs that manage how hardware works, and how that hardware communicates with other programs.
- Ask students what operating systems they are familiar with and what tasks they think are handled by an OS. Specific types of OS will be covered in the following lesson, but it is good to remind students that they already interact with a few different operating systems.

A System Has Many Purposes (30-45 minutes)

- Distribute Student Handout 3.2.1 – Purposes of Operating Systems and encourage students to fill in answers as you go through today's lesson.

Interface between Applications and Hardware

- Operating systems serve many purposes and there is some collaboration across these roles.
- At its most basic level, an operating system coordinates human requests for operations into specific instructions computer hardware can understand and perform – it provides the translation needed between user input, software commands, and hardware actions.
- It can accept user requests through two main interfaces:
    - A ***graphical user interface*** (GUI) is a visual system that allows a user to interact with an operating system using icons, buttons, and menus instead of text-based commands. A GUI is easier to use because of the visual nature of the system.
    - A ***command line interface*** (CLI) is a text-based system that allows a user to interact with an operating system using text-based commands instead of icons, buttons, and menus. A CLI allows for more control of programs and systems but provides less user feedback and requires more basic knowledge to use.

Disk Management

- ***Disk management*** is the OS task responsible for ensuring efficient use of storage resources, maintaining data integrity, and enhancing system performance. It is in charge of any task related to organizing, managing, and maintaining storage devices.
- It achieves these goals through:

- **Partitioning:** divides physical disk space into multiple logical sections which the OS treats as separate disks.
- **Formatting:** prepares a disk or partition for OS use by setting up a file system.
- **Disk allocation:** manages disk space to ensure efficient use of storage.
- **Disk scheduling:** optimizes the order in which disk operations are performed when multiple read and write requests are made.
- **Backup and recovery:** processes manage the creation of file and disk image backups to protect against data loss.

Task and Process Management

- ***Task management*** oversees and controls the execution of tasks (or processes) in a computing environment.
- ***Process management*** involves handling processes through their lifecycle.
- **Multitasking:** The operating system's ability to execute multiple tasks or processes seemingly simultaneously by rapidly switching between them.
- **Concurrency:** The efficient use of resources to execute multiple processes or threads in overlapping time periods, enabling parallelism when multiple processors are available.
- **Deadlock:** A situation where processes are unable to proceed because each is waiting for the other to release resources. The OS must detect and resolve deadlocks to prevent system halts.

Application Management

- ***Oversees the execution***, performance, and interaction of applications within a computing environment.
- **Application Lifecycle:** The OS manages the entire lifecycle of an application, from launching and running to closing and cleanup.
    - Application execution – load, execute, and terminate applications based on user initiation (ex. clicking an icon in a GUI or typing an appropriate command in a CLI)
    - Application installation and management – install, update, and uninstall applications.
- **Dependency Management:** The OS handles application dependencies, including libraries and frameworks required for application execution
- Application communication – manage inter-application communication through inter-process communication (IPC) or application programming interfaces (APIs)
- **User Profiles and Settings:** The OS supports managing user profiles and application-specific settings, allowing personalized configurations and preferences.
    - Application isolation and security – sandboxing (confining) applications into a controlled environment, enforcing permissions and access controls for applications, and using virtual machines for some applications to enhance security and avoid interference across applications

Device Management

- ***Device management*** is the controlling and coordination of hardware devices.
  - Device drivers – specialized programs that enable OS to hardware device communication
  - Device installation and configuration – recognizes and configures new devices upon connection; may be automatic or may require user interaction
  - Input/Output (I/O) management – handles data transfer between applications, devices, and the system
  - Device scheduling – apply queuing and prioritization algorithms to manage access to devices
  - Error handling and diagnostics – detect, handle, and report errors related to hardware devices
  - Device hot-plugging and dynamic management – allows users to connect or disconnect devices without restarting the system
  - Device sharing and multiplexing – allows multiple users or applications to access the same device; ex. queuing print jobs from multiple devices to one printer
  - Device configuration and management tools – provides tools to manage and configure devices, such as ‘utilities’ and ‘control panels’

Access Control

- ***Access control*** regulates access to system resources and defines user permissions.
- Controls who can access specific data or functions.
- Protects system security by enforcing access restrictions.
- Authentication and authorization determine user access levels.
  - Access Control Lists (ACLs): Set permissions for individual users or groups.

Activity: 3.2.1 – Can You Handle This? (10-20 minutes)

- Students will analyze tasks a user performs in an operating system and classify them based on the OS function responsible.
- Consider if you want students to work individually or in groups. The scenarios are written into the Student Handout as well as the Activity Slides.

1. You are creating an account for an employee on a new device and setting their privileges to "User" instead of "Admin". Which OS function is handling your request?

    Access Control

2. An employee's computer is running slowly, and the employee approaches you for help. You want to check what programs are running and close some that aren't essential to the work the employee is doing. Where should you look? Which OS function are you relying upon?

    Look in Task Manager (Windows)/Activity Monitor (macOS);

    OS function is Task and Process Management

3. You are connecting to a new printer in the office. Which OS function is responsible for checking for compatible drivers?

    Device Management

4. A new employee receives their device with default programs installed. Which OS function is in charge?

    Application Management

5. You are creating a new backup volume on your hard drive. First you need to create a partition, then format a file system. Which OS function is in charge?

    Disk Management

Putting It All Together (5-10 minutes)

- What are the main purposes of operating systems?

    Provides an environment for software and hardware to interact. Translates user requests into readable system calls and hardware-specific instructions.

- Which function is in charge of terminating a process?

    Process Management

- You notice you cannot alter a setting on your device. Which function is likely impacting you?

    Access Control

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